

Inheritance Language: Allele to Phenotype

LAUNCH • Science • 40 minutes

I can use inheritance vocabulary to connect alleles, genotype and phenotype in a simple monohybrid model.

Prepare before pupils arrive

Setup: Prepare large P/p tiles and keep the complete-dominance rule visible throughout.

Safety: Keep contexts fictional and botanical; interrupt deterministic or value-laden language calmly and precisely.

Equipment: Large P/p allele tiles, genotype cards, phenotype cards, vocabulary mat and pencil.

Safety: Use only the neutral plant model. Do not ask pupils to infer their own or relatives' genotypes or disclose family medical information.

Fallback: Point to or direct an adult to pair allele tiles and scribe the exact classification reason.

Access and participation

Offer reading aloud, pointing, signing, quiet thinking, a no-touch route and exact scribing where these support the learner. Keep the same subject decision. In shared work, let each learner commit before feedback; use a partner as card mover or checker, then swap. Avoid public speed rankings.

Source lesson curriculum link: Explain monohybrid inheritance using genetic diagrams and Punnett squares.

40-minute teaching sequence

Stage / total time	Teaching move	Adult support / response
Arrival · 4 min	Define gene. Describe how DNA and chromosomes are related.	Wait, regulate and retrieve through the lightest workable route. If recall is inaccessible, point to one retrieval sentence; do not teach today's answer.
Starter · 4 min	Think first. Point, say, sign or write your choice.	Protect curiosity: receive every first idea without judgement. Ask 'What makes you think that?' and preserve the prediction for later evidence.
I do · 5 min	allele — version of a gene genotype — allele-letter combination phenotype — observed characteristic Use one simplified pea-flower gene with two alleles: P and p. Capital and lowercase are symbols, not ratings.	Let the teacher/model carry the explanation; pupils watch, point or track. Use a visual cue only if access is the barrier, then fade it.

Stage / total time	Teaching move	Adult support / response
We do · 5 min	Commit to an answer. Show the evidence you used.	Scan every response mode. Do not infer understanding from handwriting or confidence. Secure: transfer. Mixed: contrast. Misconception: counterexample then a fresh question.
I do 2 · 4 min	Dominant describes expression in a heterozygote; it does not mean stronger, healthier, better or more common. Phenotype is not always determined by one gene alone; this pea model deliberately uses a simple monohybrid characteristic.	Model one complete evidence-to-explanation sentence, then pause. Keep the science words; change density, modality or adult role before changing the goal.
We do 2 · 4 min	Classify each card, then justify the phenotype using the simplified dominance rule. ALLELES GENOTYPE ZYGOSITY PHENOTYPE LIMIT Use the numbered cards in your pupil pack.	Protect pupil operation and thinking; support handling only where needed. Ask what changed and what stayed the same before supplying a scientific word.
Independent · 10 min	Decode six genotypes, identify phenotypes and explain the limits of the simple P/p model. Record: Accurate use of allele, genotype, phenotype, homozygous and heterozygous plus one model limitation.	Protect independence. Accept equivalent evidence routes and scribe exact meaning. Prompt only as much as needed; fade as soon as the pupil continues.
Exit · 4 min	Explain why Pp is heterozygous and gives purple flowers in this model. Point to, read or explain the evidence you made.	Genuinely receive one final response and name back the Science heard or seen. Give one next move; the pupil edits or re-attempts on the existing work.

The timing belongs to the whole stage. Several PowerPoint slides may share it; do not restart that time on every slide. Full source-stage text is in the PowerPoint speaker notes and original HTML.

Answers, evidence and feedback

Hinge answer: Pp is a heterozygous genotype with the purple phenotype.

Yes. Pp contains two different alleles, so it is heterozygous; P is expressed in this complete-dominance model.

Misconception repair

Repaired. Dominant is an inheritance relationship in the model, not a claim about value, strength or how common an allele is.

Guided checkpoint

Yes. Pp contains two different alleles, so it is heterozygous; P is expressed in this complete-dominance model. First compare the two allele symbols, then use the model's dominance rule.

PP

HOMOZYGOUS. PP is homozygous dominant. Compare the two symbols: are they the same or different?

Pp

HETEROZYGOUS. Pp is heterozygous. Compare the two allele versions.

White flowers

PHENOTYPE STATEMENT. White flowers is a phenotype statement. Ask whether this is an allele combination or an observed feature.

Model limitation

The model assumes one gene, two alleles and complete dominance; many human and organism characteristics are more complex and influenced by environment.

Independent evidence

Accurate use of allele, genotype, phenotype, homozygous and heterozygous plus one model limitation.

Supported: Match PP, Pp and pp to same/different alleles and purple/white phenotype. Keep the P/p key and icon-backed word mat visible.

Standard: Classify six genotype/phenotype statements and explain two matches. Use: ___ is the genotype; ___ is the phenotype because ___.

Stretch: Correct three dominance misconceptions and evaluate the one-gene complete-dominance model. Do not introduce codominance, sex linkage or Higher-tier drift.

Record the teaching response

What the learner actually showed	Next teaching action
Secure explanation with evidence	Reduce content prompting; offer another example or a limitation.

What the learner actually showed	Next teaching action
Partly correct / mixed	Point to the deciding evidence and invite a revision.
Access barrier	Change the reading, sensory or response format; keep the subject decision.
Missing source or observation	Keep the gap visible. Name the source or authorised check needed.

Safety and evidence boundary

Use the centre's authorised practical or fieldwork method and local risk assessment. Supplied sample data, fictional place models, card feedback and rehearsals are teaching material. They are not the learner's practical observations or proof of a qualification outcome. For portfolio use, check the exact registered unit/version, accepted evidence and centre process.

Sources and companion notes

Primary classroom source: [SCI_L_W13L1_Alleles_Genotype_Phenotype_Introduce.html](#)

The uploaded lesson is included unchanged. Text and teaching steps in the PowerPoint are editable; source diagrams are placed images. Word booklets are editable. Static PDFs cannot reproduce HTML controls; use the paper cards/tables or open the original HTML.

Verification scope: matched to the uploaded title, pathway, objective, stage sequence and 40-minute timing. Embedded workbook references are retained as source locators; this is not a new line-by-line audit of every scheme-of-work row or a centre assessment sign-off.